

The Effect of D(+)-maltose and Curry on Glycogen Phosphorylase b

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Abstract:

Glycogen phosphorylase b is an inactive enzyme that is involved in the glycogen metabolism. Fluorescence spectroscopy was applied on the enzyme to identify molecular components of glycogen phosphorylase b. Its molecular components contain dimer of two identical subunits. It does not contain a phosphate group which makes it different from glycogen phosphorylase a (the active form). Glycogen phosphorylase b is commonly found in the liver, where glucose gets stored as glycogen. The experiment was done on glycogen phosphorylase b to analyze its reaction toward d(+)-maltose and Curry. To analyze its activity, we measured the absorbance of an inorganic phosphate. D(+)-maltose and Curry were used at different concentrations to find out if d(+)-maltose and Curry act as an inhibitor or activator by analyzing the change in Michaelis-Menten curve. After analyzing the data that was collected, both d(+)-maltose and Curry act as an inhibitor.

Background:

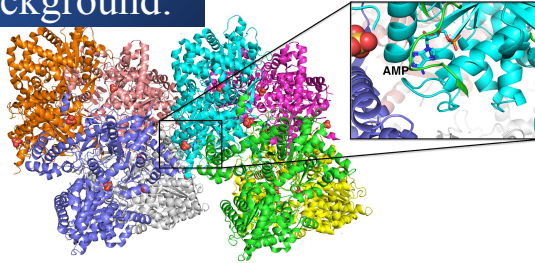


Figure 1: AMP binds to glycogen phosphorylase b and acts as an activator ().

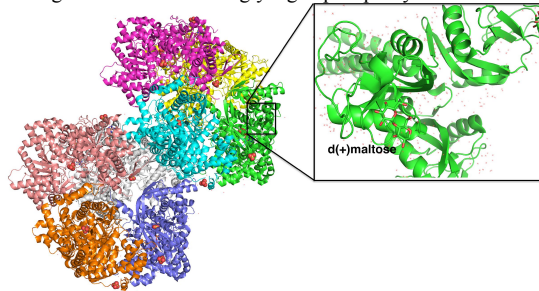


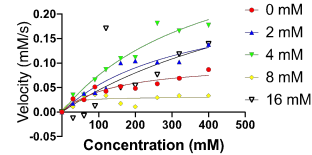
Figure 2: D(+)-maltose binds to glycogen phosphorylase b and act as an inhibitor.

Aim of Study:

To study the affection of d(+)-maltose, AMP and Curry on glycogen Phosphorylase b

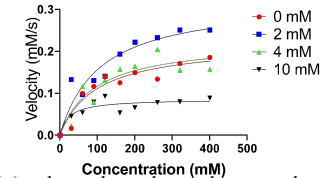
Results:

Effect of AMP on Glycogen Phosphorylase b



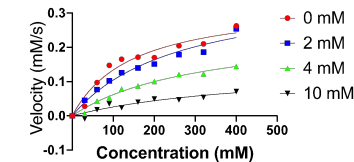
Graph 1: AMP speeds up glycogen phosphorylase b activity.

Effect of d(+)-maltose on Glycogen Phosphorylase b



Graph 2: D(+)-maltose slows down glycogen phosphorylase b activity.

Effect of Curry on Glycogen Phosphorylase b



Graph 3: Curry slows down glycogen phosphorylase b activity.

Conclusions:

Glycogen phosphorylase b is involved in the glycogen metabolism that gets regulated by three allosteric regulators (AMP, d(+)-maltose and Curry).